

CZ-1 DL Alzheimer's

Final Report (Research Paper)

CS 4850 - Sections 02 & 04 - Fall 2025

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GitHub: <https://github.com/CZ1-Alzheimers/Using-Machine-Learning-to-Diagnose-Alzheimers/tree/main>

Number of lines of Code	1100
Number of Components	4 Total Components: - Python Script - Full Dataset (ADNI/Oasis/AIBL) - SPM - CAT12
Total Man Hours (Estimate / Actual)	110 (Estimate) / 147.5 (Actual)
Status	100% complete and working as designed

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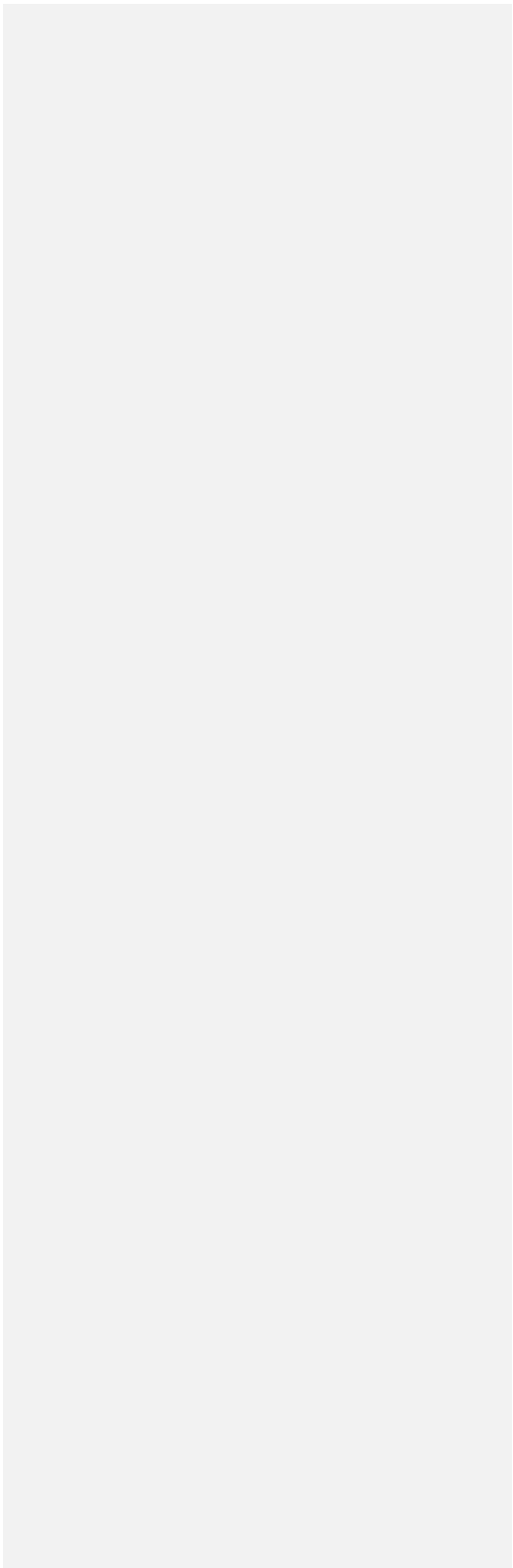
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Abstract

Early detection of Alzheimer's disease (AD) remains a significant clinical challenge, as the changes associated with cognitive decline are often subtle and difficult to identify through visual assessment alone. This study investigates modern machine learning (ML) methodologies to improve the prediction of cognitive impairment using volumetric magnetic resonance imaging (MRI)-derived region-of-interest (ROI) features. We aim to classify normal control (NC), mild cognitive impairment (MCI), and AD patients. We constructed three binary classifiers [NC vs. AD, MCI vs. AD, and NC vs. MCI] and evaluated various algorithms, including logistic regression, random forests, neural networks, and support vector machines (SVM). Using measurements generated from eight anatomical brain templates, our models learned patterns indicative of normal cognition, mild cognitive impairment, and AD. Among all tested approaches, the radial basis function (RBF) SVM consistently achieved the highest performance, reaching accuracies of approximately 70–80% depending on the classification task. We discuss the implications of this model's dominance for future clinical applications and the continued development of ML driven diagnostic tools.

Introduction

AD is a neuro-degenerative disorder which is responsible for causing 60-80% of all dementia cases throughout the United States, according to the University of Newcastle (Amir and Suhuai 2). The earlier that this disease is detected, the more effective treatment will be.

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Our proposed method is to utilize three SVM binary classifiers (NC v. AD, NC v. MCI, AD v. MCI) trained on three databases of MRI data: Alzheimer's Disease Neuroimaging Initiative (ADNI), The Open Access Series of Imaging Studies (OASIS), and Australian Imaging Biomarkers and Lifestyle Study (AIBL). We processed each image in our dataset using MATLAB, Statistical Parametric Mapping (SPM), and Computational Anatomy Toolbox (CAT12). We believe our justification for choosing a SVM model is feasible, and could give insight into potential scientific purposes for other ML models

Related Work

Since ML tools have rapidly improved their pattern recognition abilities, more scientists are looking to use this technology to recognize patterns too subtle for the human eye and possibly come to conclusions no human scientist could reasonably come to. In the last five years, there have been a plethora of papers which propose utilizing an AI model to analyze volumetric brain data, be it through MRI or PET scans, and form an understanding of the commonalities between brains that are NC, have MCI, and have AD.

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Researchers at the University of Newcastle explore interesting new Convolutional Neural Network (CNN) Architecture (ConvADD), which unparalleled accuracy and F1-scores when compared to the popular models (Mohammed et al. 311). Researchers out of McMasters University also studied the use of CNN's for the purpose of AD disease detection, yet they compared the benefits and drawbacks of using a 2D CNN vs a 3D one (Ebrahim and Luo 11). A team. Steering away from Deep Learning, a team of independent researchers also took a wide look at various technologies used to accomplish the AD Classification task (Shahbaz et al. 4)

Many of these methods involve utilizing computer vision technology to segment the brain, only then using binary classifiers to form profiles of NC, MCI, and AD brains. While authors such as Amir Ebrahimi and Suhui Luo note techniques like this are novel, we believe that enhancing our binary classifiers should be the primary focus of our study and therefore decided to use a standardized MRI pre-processing pipeline.

Neuroimaging and Preprocessing

Detection of AD is typically based on a variety of clinical assessments, medical consultations, and the examination of brain scans. Researchers from Chosun University in Korea note that MRIs are the most widely utilized biomarker, employed in over 80% of single-modal AD detection studies (Nagaraj et al. 1). MRIs offer high spatial resolution necessary for analyzing anatomical structures of the brain. However, accurately detecting AD using algorithms is challenging due to inherent complexities in brain anatomy, low image quality, and difficulties associated with preprocessing and segmentation steps (Nagaraj et al. 2). Effective diagnostic systems rely on strict preprocessing pipelines to handle discrepancies and variability across a wide variety of subjects and scans (Nagaraj et al. 4).

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Accurate diagnosis often depends on the quantitative analysis and segmentation of specific brain structures. Brain image segmentation aims to divide images into defined regions like gray matter, white matter, and cerebrospinal fluid. Structural abnormalities, particularly tissue atrophy, are popular indicators used in diagnosing AD. Specifically, atrophy of medial temporal structures is considered a valid diagnostic marker even at the MCI stage. The hippocampus and the precuneus are identified as highly sensitive biological indicators of AD, particularly in the early stages.

Classification Algorithms

Traditionally, the detection of AD using ML has employed a wide range of algorithms. Common classification algorithms in early ML research included Neural Networks (NN's) and SVMs applied to MRI scans. Traditional, established methodologies such as SVM still maintain relevance in AD classification studies; however, the field has seen a shift toward advanced deep learning methods since the rapid evolution of deep learning algorithms.

As for newer deep learning models, Convolutional Neural Networks (CNNs) have revolutionized outcomes when it comes to disease detection. CNNs merge feature extraction, feature selection,

and classification all into a single process. The use of CNNs is prominent, with the *Journal of Medical Imaging* accounting for about 70% of AD detection methods utilizing CNN systems. CNN's have proven more efficient to train as they include the segmentation process within the pipeline, not all medical communities have the infrastructure or environment to dedicate hundreds of hours in compute time and thousands of watts towards new technology to accomplish a task that MATLAB and SPM have already perfected the pipeline for.

Datasets and Classification Targets

The ADNI dataset is the most extensively utilized database in this research domain. ADNI contains comprehensive information, including MRI, PET, and clinical data, aimed at identifying and validating biomarkers for tracking AD progression. The OASIS dataset is also commonly applied for brain MRI segmentation and AD diagnosis.

The classification objective often extends beyond simple AD versus NC distinctions to include multi-class categorization. Given that MCI is recognized as the prodromal phase of AD, efforts focus heavily on differentiating between NC, MCI, and AD states to facilitate early detection.

Methodology

Method Overview

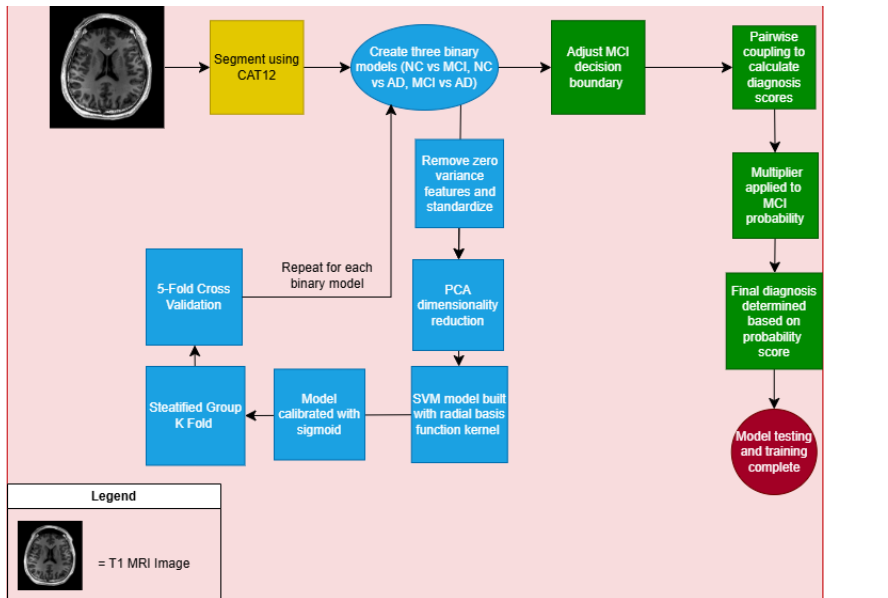


Figure 1- Methodology Diagram

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Method Explanation

Each component in our methodology serves a vital purpose to accomplishing our goal. The following pipeline was implemented to create, train, and test our model.

Segmentation using CAT12 of Input MRI Image

The purpose of this component is to convert raw T1 MRI images into usable imaging features using SPM and CAT12. The SPM and CAT12 perform structural MRI processing and tissue segmentation. CAT12 provides a fully automated pipeline for structural measurements of the brain that integrates tissue classification, bias-field correction, spatial normalization, and modulated gray-matter maps.

The CAT12 segmentation is done using SPM's probabilistic generative model to estimate tissue classes and correct nonuniformities. Gaussian mixture distributions are used to model voxel intensities and parameter estimation is performed with an expectation-maximization optimization procedure. CAT12 enhances the segmentation accuracy with a three-class partial volume model to refine gray matter and white matter boundaries. The refined gray matter maps are used for atlas-based feature extraction using eight different templates. The eight templates used are: Automated Anatomical Labeling (AAL), AAL2, AAL3, Brainnetome 246 (BN), Brodmann, Hammersmith, Harvard-Oxford Cortical and Subcortical (HOCS), and Jülich Cytoarchitectonic (JC).

The AAL atlas template divides the cerebrum into 116 defined regions based on a manually labeled single-subject MRI. It is frequently used to map structural and functional neural connections due to its broad anatomical coverage. The AAL2 and AAL3 are updated versions of the AAL template with improved boundaries, expanded structural coverage, and finer labeling for more advanced analyses. AAL2 and AAL3 produce 120 and 166 regions respectively.

The BN atlas template provides high-resolution and connectivity information divisions with 246 regions derived structural and functional connectivity profiles. It offers precise descriptions of the cytoarchitectonic and connectomic subdivisions of the brain.

The Brodmann atlas template describes cortical organization based on differences in neuronal layering. It is not as precise with spatial precision compared to other templates, but is widely used to map structural and functional results cortical areas. It provides 82 features.

The Hammersmith atlas is a manually segmented, multi-subject atlas that provides information for 83 cortical and subcortical regions. It provides high anatomical accuracy for structural volumes in high resolution MRI images.

The HOCS atlas template provides probabilistic atlases from various manually segmented MRI images. It has a max-probability threshold of 25% and yields discrete and reliable labels across 1mm isotropic space. HOCS provides information for 112 regions.

The JC atlas template is created from post-mortem histological analysis that maps cortical and subcortical regions. It is used to determine voxel-wise labels for feature extraction and produces information for 69 regions.

After the image is segmented, the data is mapped using all eight templates, creating a total of 914 features. Features produced by multiple templates are merged and only counted once. These features represent various aspects of the image, such as regional measurements or intracranial volume, and can be interpreted by the model. The segmentation extracts meaningful features, and the templates ensure that features across all images are represented in the same way.

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Creating Three Binary Models

Three binary models are used to make comparisons between each diagnosis label (NC vs AD, MCI vs AD, NC vs MCI). We use three binary models due to the ambiguous nature of an MCI diagnosis. Because MCI often resembles NC, predicting it can be difficult. Binary models allow for flexible processing when determining the final diagnosis. They also are better at determining boundary cases between each diagnosis. Each model is trained with identical pipelines, but different hyperparameters due to different needs between models.

Removing Zero-Variance Features and Standardization

Features with no variance or unique information are dropped to reduce noise. Equation 1 (Reinhold, Jacob C et al) is performed to convert each nonzero feature to have zero mean and unit variance to allow unique features to be comparable. x_j represents the original feature value, u_j is the mean of feature j across all subjects, and σ_j is the standard deviation of feature j across subjects. $\tilde{x}_j$ represents the normalized feature.

$$\tilde{x}_j = \frac{x_j - u_j}{\sigma_j}$$

Equation 1- Intensity Normalization

This standardization is important before principal component analysis (PCA) and the model are applied to the data so that distances are scaled and components are not skewed. Applying standardization to the dataset cleans the data and prepares it for feature scaling.

PCA Dimensionality Reduction

PCA is applied to reduce the highly dimensional dataset into a more manageable, less dimensional dataset. Each binary model has a PCA value of 200 (NC vs MCI and NC vs AD) and 150 (MCI vs AD) that is chosen based on a grid search. The dataset for that model is then reduced from 914 features to the determined PCA value. This reduction in dimensionality makes the differences between each label more obvious while preserving variance in the dataset.

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This is performed by computing the correlation matrix of standard features, performing eigen decomposition, and projecting data onto the leading principal components. The top k components are kept to represent a fraction of variance, where k is the chosen PCA value.

SVM Model with Radial Basis Function Kernel

Each binary model is built as an SVM model with a radial basis function (RBF) kernel. An SVM model is used because they are good at creating nonlinear decision boundaries between classes. The RBF kernel maps the inputs into a feature space with infinite dimensions so that complex boundaries can become linear using Equation 2 (Chen, Sheng). In this equation, x and x' represent two feature vectors in the input space. γ represents a positive scaling parameter that controls the width of the kernel. The exponential functions the distance into a similarity measure between 0 and 1.

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

Equation 2 - Gaussian Kernel

The decision function utilizes Equation 3 (Boser, Bernhard E., et al), where x is the input vector being classified, x_i is the i -th training sample, and y_i is the class label. The variable a_i is the Lagrange multiplier, with nonzero values indication support vectors. $K(x_i, x)$ is the Gaussian kernel function from Equation 2. The variable b represents the bias term that shifts the decision boundary.

$$f(x) = \sum_{i=1}^n a_i y_i K(x_i, x) + b$$

Equation 3 - SVM Decision Function

This type of model is good for this purpose due to its ability to handle nonlinear data with moderate dimensionality. It is well equipped to handle a small dataset, such as ours.

Calibrate Model with Sigmoid

Calibration via sigmoid is helpful to convert the decisions from SVM models into probability scores. These probability scores can be combined across each model, which is necessary to predict a final diagnosis. The sigmoid function maps decision scores to class probabilities using Equation 4 (Cox, D. R.). $P(y = 1 | x)$ represents the probability of output variable y equaling the positive class in a binary comparison, where x is the input variable. $f(x)$ describes the function being transformed into a probability. A and B are learned scaling parameters found by fitting the sigmoid to a validation set.

$$P(y = 1 | x) = \frac{1}{1 + \exp(Af(x) + B)}$$

Equation 4 - Sigmoid Function

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Many operations further in the model's pipeline require probabilities in place of raw SVM decision scores, so calibration is a vital step to ensuring accurate predictions.

Stratified Group K-Fold

The purpose of Stratified Group K-Fold is to split data across training and testing folds while ensuring that data from the same subjects stay in the same group. This prevents inaccurate high-performance predictions from the model interpreting data it has seen before.

The stratification portion of this component ensures that each fold has approximately similar class proportions. This step ensures that folds and class evaluations are fair and balanced, while avoiding data leakage.

5-Fold Cross Validation and Confusion Matrix

5-fold cross validation is a technique used to tune hyperparameters and evaluate the performance on each binary model. This works by taking the five folds generated by the Stratified Group K-Fold component and dividing them into four training folds and one validation fold. This process is repeated five times, one for each fold.

This process produces an evaluation of the model based on the average performance across all five folds. Overall accuracy, sensitivity, specificity, area under the curve (AUC), and F1 scores are evaluated and stored. Receiver operating characteristic (ROC) curves are also evaluated and plotted.

This component serves to reduce variance in model evaluation and determine the best hyperparameters to use in each model via grid search based on the calculated metrics. The hyperparameters it tunes are PCA number, selector k, model C, model gamma, and custom class weights to boost MCI predictions.

Because accurate predictions are vital to our goal, we utilized a confusion matrix (CM) to evaluate how often true and false predictions occur for each diagnosis label. For each binary model, a mean CM was calculated and displayed to help us determine weak points in each binary model.

After predicting the diagnosis on our testing set, made with 20% of our dataset, a 3x3 CM is calculated and displayed. The CM displays true NC, MCI, and AD predictions compared to false predictions, and displays how many times one label was confused for another. Overall accuracy is then calculated as well based on the number of correct predictions in the CM.

Pairwise Coupling to Calculate Diagnosis Scores

When the models have finished training, predicting the final diagnosis on an MRI image is the next goal. To accomplish this, we utilize pairwise coupling to combine binary probabilities and convert them into multiclass probabilities. The multiclass probability vector is represented as Equation 5, where $\mathbf{p}$ represents the probability vector, p_{NC} is the probability of the subject being

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NC, p_{MCI} is the probability of the subject having MCI, and p_{AD} is the probability of the subject having AD.

$$\mathbf{p} = (p_{NC}, p_{MCI}, p_{AD})$$

Equation 5 - Probability Vector

We utilized the Wu-Ling-Weng pairwise coupling algorithm to perform this task. The algorithm works by assigning pairwise probabilities from each classifier, then determining the class probability that best identifies the image using Equation 6 (Wu, T.-F., C.-J. Lin, and R.-C. Weng). In Equation 6, p_i represents the unknown probability of class i , p_{ij} represents the pairwise SVM probability of class i winning over class j , and K represents the total number of classes. $\frac{p_i}{p_i + p_j}$ represents the global probability of class i winning over class j . $F(\mathbf{p})$ is the total mismatch error between pairwise and global probabilities.

$$\min_{\mathbf{p}} F(\mathbf{p}) = \sum_{i=1}^K \sum_{j=i+1}^K (p_{ij} - \frac{p_i}{p_i + p_j})^2$$

Equation 6 - Wu-Ling-Weng Equation

This technique is used because the generated probabilities are more consistent and more easily interpreted compared to inconsistent voting algorithms.

Adjusting MCI Decision Boundary and Applying MCI Multiplier

A common issue with this and other similar models is MCI prediction. MCI, by nature, closely resembles NC and has more subtle features. Because of this, MCI prediction would be disproportionately low compared to NC and AD prediction. To fix this, adjust the MCI decision boundary and apply a multiplier to the MCI probability.

By adjusting the decision boundary, the threshold for what the model considers MCI is lowered, making predictions more likely. Applying the multiplier also increases the overall probability of an MCI diagnosis. Through testing, we determined that shifting the decision boundary by 0.18 and declaring a small multiplier of 1.01x produced the best results.

These changes in MCI probability give the diagnosis label the boost it needs to be accurate without sacrificing AD and NC diagnosis accuracy.

Final Diagnosis based on Probability Score

This final component converts the multiclass probabilities [into](#) a final determined diagnosis label for clinical use. All label probabilities are compared after MCI boosting is complete, and the label with the highest probability score is determined to be the final diagnosis. Experimental Results

Data and Experiment Settings

The methodology above was implemented using a dataset of multiple MRI images containing subjects from ADNI, OASIS, and AIBL databases. Each image is processed through CAT12 for segmentation, then saved with all features in a single dataset file. Each image is then labeled with its proper diagnosis. The dataset is then randomly split into 80% training and 20% testing, with Stratified Group K-Fold ensuring that images from the same subject are grouped together. Our complete dataset consists of 1551 images, with 520 images from the ADNI dataset, 211 from AIBL, and 820 images from OASIS. The dataset consists of 602 NC images, 408 MCI images, and 541 AD images.

During model training, Stratified Group K-Fold is utilized to prevent data leakage and keep images from the same subject in the same fold. An outer loop of five folds is used to estimate performance, while an inner loop of three folds is used to tune hyperparameters.

Three binary classifiers (NC vs MCI, NC vs AD, MCI vs AD) are made and evaluated. These classifiers are used to build a pairwise system that is then combined using the above methodology to perform multiclass probability estimates.

Implementation Details

All MRI feature vectors move through the same preprocessing pipeline. Zero-variance features are removed to clean the data, then standardization is applied to convert all features to zero mean and unit variance. Dimensionality is then reduced with PCA. The number of dimensions it is reduced to is determined by each classifier during hyperparameter tuning.

After preprocessing, each classifier is modeled using an SVM with an RBF kernel. Hyperparameters are optimized using grid search and cross validation. Synthetic Minority Over-Sampling Technique (SMOTE) oversampling is also applied to each training fold to aid with class imbalance. When training is complete, sigmoid based calibration is used to convert the classifiers into probability predictors.

The three calibrated binary probability predictors are combined with pairwise coupling to determine the overall most probable diagnosis for any given image. MCI decision boundary shifting and multipliers are applied to aid with MCI ambiguity.

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Major Results

As seen in Table 1, each binary classifier performs well with metrics being in their predicted range.

Comparison	Accuracy	AUC	Sensitivity	Specificity	F1-Score
NC vs MCI	0.667872	0.722778	0.395709	0.859942	0.477875
NC vs AD	0.804809	0.885175	0.763742	0.841388	0.787457
MCI vs AD	0.704832	0.753465	0.772687	0.617057	0.747717

Table 1- Complete Dataset Binary Classifier Comparisons

Table 2 describes the performance of the model using the ADNI dataset only. The ADNI dataset contains 161 NC images, 255 MCI images, and 104 AD images.

Comparison	Accuracy	AUC	Sensitivity	Specificity	F1-Score
NC vs MCI	0.594687	0.630516	0.874015	0.182049	0.72151
NC vs AD	0.877117	0.940627	0.785753	0.942007	0.823271
MCI vs AD	0.703674	0.628071	0	0.990909	0

Table 2 - ADNI Dataset Binary Classifier Comparisons

Table 3 describes the performance of the model using the OASIS dataset only. The OASIS dataset contains 358 NC images, 86 MCI images, and 376 AD images.

Comparison	Accuracy	AUC	Sensitivity	Specificity	F1-Score
NC vs MCI	0.791138	0.553611	0.108117	0.958229	0.164327
NC vs AD	0.785208	0.874623	0.767999	0.814326	0.784094
MCI vs AD	0.818825	0.654675	0.986991	0.085735	0.898223

Table 3 - OASIS Dataset Binary Classifier Comparisons

Table 4 describes the performance of the model using the AIBL dataset only. The AIBL dataset contains 83 NC images, 67 MCI images, and 61 AD images.

Comparison	Accuracy	AUC	Sensitivity	Specificity	F1-Score
NC vs MCI	0.620652	0.696369	0.250319	0.90658	0.328571
NC vs AD	0.886957	0.969585	0.78083	0.975	0.851397
MCI vs AD	0.656667	0.783807	0.613939	0.724127	0.622076

Table 4 - AIBL Dataset Binary Classifier Comparisons

Of all three classifiers, NC vs MCI performs the worst. This is due to the high similarity between NC and MCI images and MCI's ambiguous nature. Our goal for the NC and MCI comparison was at least 65% accuracy, which the model achieved. Though the F1-score is relatively low, the accuracy and AUC scores meet expectations signifying that model is useful and usable.

The goal for the NC vs AD and MCI vs AD comparisons was to reach at least 70% accuracy, which both models achieved. MCI vs AD has the same issue as NC vs MCI, which is MCI's ambiguous nature. When compared to AD, however, the results are more accurate due to more obvious differences between MCI and AD.

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NC vs AD performed the best overall, achieving 80% accuracy. This classifier was able to perform the best due to the drastic differences between NC images and AD images.

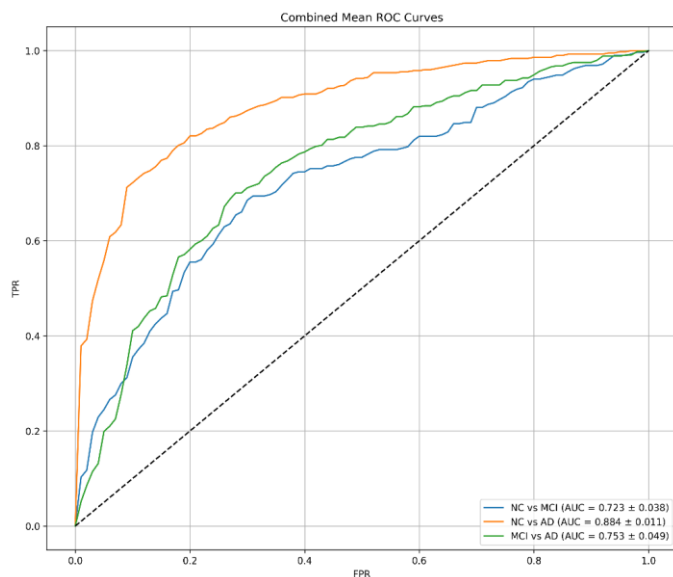


Figure 2- Classifier ROC Curves

As seen in Figure 1, each classifier performs well above the base 0.5 AUC rate. NC vs MCI and MCI vs AD have similar curves due to MCI's nature. The NC vs AD model predictably performs much better than models involving MCI due to stark differences in images and reduced ambiguity.

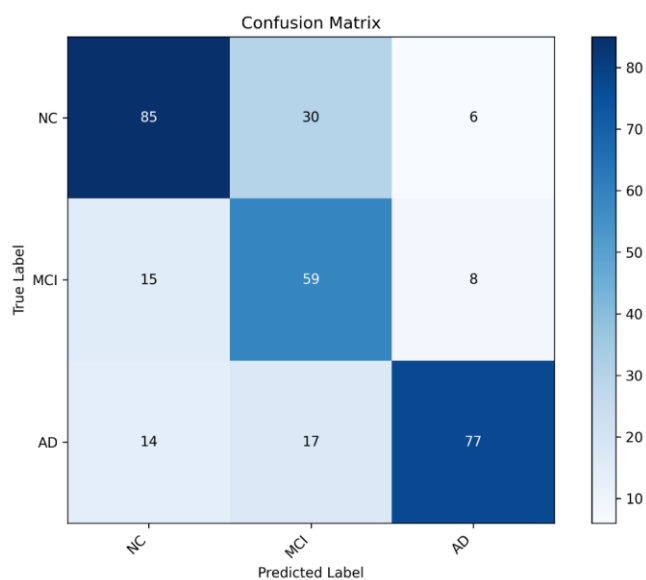


Figure 3 - Testing Confusion Matrix

Figure 2 shows the confusion matrix of the testing data. NC and AD are easiest to diagnose and are most often confused for MCI. MCI sits between NC and AD, so it understandable that ambiguity would cause it to be falsely predicted. NC is most often falsely predicted as MCI because of their strong similarities, while AD is confused as NC or MCI an almost equal amount. Probability adjustments for MCI increased its prediction accuracy, but may have had some minor negative impacts on NC detection.

Based on the CM in Figure 2, the overall accuracy of the model is 71%.

Discussion

Our goal for this project was to reach at least 70% overall accuracy when predicting diagnosis labels of MRI images. We were able to reach this goal using an accurate, calibrated SVM model.

Through our research, we determined that separating each comparison into three binary models was highly effective and made creating decision boundaries between labels much stronger. Pairwise modeling and coupling was highly effective for separating each label, then combining their probabilities to calculate a final diagnosis label.

We also determined that MCI remains highly ambiguous, which was expected prior to experimentation. Adjusting the boundary and boosting MCI's probability significantly helped detection without compromising NC or AD detection. Because we used a standard SVM model, MCI prediction accuracy for our model can likely not be improved without implementation of more complicated models.

Overall, our model demonstrates strong potential to diagnose AD in clinical patients. While our model is not accurate enough to completely declare a diagnosis, it could serve as a way to support clinicians catch AD and MCI development early. Predictions of MCI and AD in patients through our model would encourage clinicians to take second looks at MRIs or reconsider factors that might have been missed in patients thought to have no impairment.

References

1. Ebrahimi, Amir, and Suhuai Luo. "Convolutional neural networks for Alzheimer's disease detection on MRI images." *Journal of Medical Imaging*, vol. 8, no. 2, 29 Apr. 2021, p. 024503. DOI: 10.1117/1.JMI.8.2.024503.
2. Shahbaz, Muhammad, et al. "Classification of Alzheimer's Disease using Machine Learning Techniques." *In Proceedings of the 8th International Conference on Data Science, Technology and Applications (DATA 2019)*, 2019, pp. 296-303.
3. Sarraf, Saman, and Ghassem Tofighi. "Classification of Alzheimer's Disease Structural MRI Data by Deep Learning Convolutional Neural Networks." *arXiv preprint arXiv:1607.06583*, 2016.
4. Miltiadous, Andreas, et al. "DICE-Net: A Novel Convolution-Transformer Architecture for Alzheimer Detection in EEG Signals." *IEEE Access*, vol. 11, 12 July 2023. DOI: 10.1109/ACCESS.2023.3294618.
5. Yamanakkanavar, Nagaraj, Jae Young Choi, and Bumshik Lee. "MRI Segmentation and Classification of Human Brain Using Deep Learning for Diagnosis of Alzheimer's Disease: A Survey." *Sensors*, vol. 20, no. 11, 7 June 2020, p. 3243. DOI: 10.3390/s20113243.
6. Alsubaie, Mohammed G., Suhuai Luo, and Kamran Shaukat. "ConvADD: Exploring a Novel CNN Architecture for Alzheimer's Disease Detection." *[Source title not explicitly provided in excerpt]*, vol. 15, no. 4, 2024.
7. El-Assy, A. M., et al. "A novel CNN architecture for accurate early detection and classification of Alzheimer's disease using MRI data." *Scientific Reports*, 4 Feb. 2024. DOI: 10.1038/s41598-024-53733-6.
8. Reinhold, Jacob C et al. "Evaluating the Impact of Intensity Normalization on MR Image Synthesis." *Proceedings of SPIE--the International Society for Optical Engineering* vol. 10949 (2019): 109493H. doi:10.1117/12.2513089

9. Chen, Sheng. ‘Orthogonal-Least-Squares Forward Selection for Parsimonious Modelling from Data’. *Engineering*, vol. 01, no. 02, Scientific Research Publishing, Inc, 2009, pp. 55–74, <https://doi.org/10.4236/eng.2009.12008>.
10. Boser, Bernhard E., et al. ‘A Training Algorithm for Optimal Margin Classifiers’. *Proceedings of the Fifth Annual Workshop on Computational Learning Theory*, Association for Computing Machinery, 1992, pp. 144–152, <https://doi.org/10.1145/130385.130401>. COLT ’92.
11. Cox, D. R. ‘The Regression Analysis of Binary Sequences’. *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 20, no. 2, 2018, pp. 215–232, <https://doi.org/10.1111/j.2517-6161.1958.tb00292.x>.
12. Wu, T.-F., C.-J. Lin, and R.-C. Weng. "Probability Estimates for Multiclass Classification by Pairwise Coupling." *Journal of Machine Learning Research*, vol. 5, 2004, pp. 975-1005.

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